

AMR ABDALLAH

📍 603-369 London Road, Sarnia ON - N7T4W5



relatora@gmail.com



519 466 8200



[Website](#)

SUMMARY OF QUALIFICATIONS

Applied AI Engineer & Software Engineer building intelligent, autonomous systems that turn complex business problems into reliable, production-ready solutions. Specializing in **AI agents, LLM-powered applications, tool orchestration, multi-agent workflows, and modern software architecture**, with hands-on expertise spanning AI, cloud, backend, frontend, and distributed systems.

I bridge the gap between emerging AI capabilities and practical business execution—designing and integrating systems using **LLM APIs, Model Context Protocol (MCP), function/tool calling, GitHub Copilot Studio, local LLMs, and AI-assisted software development**. Experienced in taking AI concepts from architecture and experimentation through integration, deployment, and production.

CORE TECHNICAL EXPERTISE

- **AI & Agentic Systems:** Autonomous AI Agents, Multi-Agent Orchestration, LLM Integration, Tool/Function Calling, MCP, Prompt Engineering, GitHub Copilot Studio, Local LLMs
- **Backend & Cloud:** Go (Golang), Node.js, Python (FastAPI/Flask), Java (Spring), .NET Core, PHP (Laravel), REST APIs, Cloud Architecture, AWS, Azure
- **Frontend & Web:** React, Vue.js, JavaScript/TypeScript, HTML5/CSS3
- **Data & Systems:** PostgreSQL, MySQL, Oracle, DB2, MS SQL, MongoDB, Firebase, Redis, Kusto, Supabase, Vector Databases
- **IoT & Integration:** MQTT, Modbus, API Integration, Distributed Systems

EXPERIENCE

2021 (April) –Current Time– Full Time

Senior Software Engineer and Project Manager, AlumaPower

- Design, architect and implement databases for various internal desktop apps, mobile apps and web apps required for internal operations.
- Design and develop predictive AI models to analyze experimental and operational data, identify patterns, and support data-driven predictions and decision-making.
- Designed and implemented an AI agent using Microsoft Copilot Studio and Azure to secure and control access to an Azure API, creating an intelligent layer over the API pipeline for controlled and secure interaction with backend services.
- Implement among a team web apps and desktop apps for various internal operations that saves the team thousands of hours spent doing analysis manually as well as many internal tools and software that tie pipelines together.
- Coordinate teams to accomplish projects of varying size and complexity.
- Design and implement unit tests, regression tests and end to end tests to AlumaPower's solutions.
- Assemble requirements from users, record them, and create logical and physical specifications.
- Analyze the performance and dependability of designs and to optimize architecture, develop data, process, and network models.
- Develop, install, integrate, and run computer-based systems, including mobile applications, according to a plan and schedule.
- Operating systems, communications environments, and application software maintenance methods are evaluated, tested, troubleshoot, documented, upgraded, and developed.

- The development of software and integrated information systems, process control software, and other embedded software control systems may be led and coordinated by teams of information systems professionals.
- Technical writing and documentation of software.

2021 (April) – 2022 (July) – Full Time

IT Project Manager, Industrial Web Apps

- Coordinate teams to accomplish projects of varying size and complexity.
- Coordinate teams to accomplish projects of varying size and complexity.
- Organize the team, time and resources to make sure the projects meet stated requirements and are completed on time.
- Provide direction to the project team for all phases of development in the project lifecycle.
- Supervise the timely completion of software projects, providing technical expertise and leadership abilities to resolve issues.
- Work proactively to identify obstacles within the project teams and work creatively to propose solutions.
- Lend experience and guidance to project conception and feasibility analyses.
- Maintain backlog for development projects and prioritize existing and incoming tasks.
- Work closely with Software Development team and Deployment team to deliver simultaneous projects if needed.
- Developing and Peer Reviewing multiple projects.

2021 (Sep) – Current Time – Part Time

Professor, Lambton College

Teaching Front-End-Web Development I covering HTML5 and CSS3 course, as well as, teaching Front-End-Web Development II covering JavaScript, jQuery and an introduction to Vue and React.

2020 (Nov) – 2021(May) – Full Time

Full-stack Developer, Eye-Look Media

Working with a team developing a digital signage solution portal that manages client's media across their media kiosks. A client like Canadian Tire can intelligently distribute and control media displayed on their screens across their branches. We use a **Laravel, Vue, MySQL** stack to deliver a highly creative platform. Developing a client access progressive web app (PWA) for communication using WebSocket together with several **NodeJS** micro services that perform various tasks such as recursive file parsing, socket bridging and various automation.

2020 (May) – 2020 (Aug) – Co-op

Software Developer, Avaros EMR

Worked on configuring a complex development environment on Linux. I was part of a team that worked on building patients online booking frontend using **ReactJS, React Hooks, RxJS** and **TypeScript** creating various endpoints and building numerous components. I integrated Twilio's SMS webhooks into the scheduler backend using **Go Lang** and created the corresponding migrations in **PostgreSQL**. Worked on numerous bug fixes and functionality additions to Oscar EMR using **Java, JSP, CSS, MariaDB** and **JavaScript**. Created a highly interactive 138-page document detailing **WebSocket** server logic flow on Google docs.

2019 – present – contracts

Researcher, Lambton College Applied Research

- Researched [OrderEase](#) which is an Amazon-like business. Formulated a document for potential investors and another analyzing the strengths and weaknesses of their current paradigm and software architecture.
- Implemented a **digital signage solution** for [EyeLook Media Inc.](#) that can operate on Raspberry Pi 4. This solution gets a payload from a server endpoint then downloads the necessary assets and dynamically

generates HTML/JS/CSS that would render on the screen minding special date and time rules provided by the payload. The solution architecture was based on microservices.

- Implementing an internal system using MongoDB, Express JS, React JS, Node JS (**MERN**) stack solution for [Procor](#) to handle their day-to-day railcar operations.
- Developing key components among a team for a Hockey team React Native mobile app.
- Creating a full-stack project for a Lambton College Research Center client using NodeJS, ExpressJS, ReactJS and MySQL DB. The project involves a timeline structure displaying product rentals on a daily, weekly and monthly basis.
- Engineering a solution for [Just Vertical](#). A company that specializes in vertical hydroponics. I am responsible for creating a **Flutter** app that communicates via WebSocket with an AWS Lambda **microservice** that in turn fires and receives **WebSocket** traffic to a **Quasar/Electron** app (VueJS) installed on a **Raspberry Pi** embedded in the units. The app can schedule automated tasks like controlling irrigation and light while dashboarding humidity and temperature readings. I Engineered and implemented this solution.

Software Developer, Hired by Tertec

Worked on an Internet of Things (IoT) project. Using Vue JS, D3 JS and Rete JS to build interactive components that can be dragged, dropped and manipulated.

Tutoring, Lambton College

Helped students with courses such as Python, Database Development, Java SE/EE, C#, JS, Program Logic and Design and NodeJS.

2016 – 2018

IGCSE ICT Teacher, Modern Education Schools

Gathered resources and created the syllabus that was based on Microsoft Word, Excel, PowerPoint and Access together with Scratch Programming, flow charts, HTML and CSS for web development. Built numerous internal solutions for the school including a grading system.

2006 – 2016

American System ICT Teacher, Modern Schools of Egypt

Gathered resources and created the syllabus that was based on Microsoft Word, Excel, PowerPoint and Access together with Scratch Programming, flow charts, HTML and CSS for web development. Built numerous internal solutions for the school.

2003 – 2006

Software Developer, Gulf Network Solutions

- SFA (Sales Force Automation): A developer to Sales Force automation solution using Pocket PC as the sales force terminal, MS SQL replication.
- McDonalds Egypt E-Learning Card Slips: Implementing the logic and algorithm of randomizing and categorizing of the E-Learning card slip batches that McDonald's has issued for promotion.
- Other utilities, services and tools for GNS internal usage.

2000 – 2003

Software Developer, Middle East Network Solutions

- Developed as part of a team a SMS based banking solution for the National Bank of Egypt.
- Worked on an automated call center solution at Misr for Custody and Stock Depository. Team leader for developing their dynamic call center and instructed a team to fully adapt the solution to their system.
- Developed Goldi's dynamic call center solution.

EDUCATION

2019 (Jan)–2020 (Aug)

Computer Software & Database Development, Lambton College, Sarnia, ON

1st term GPA 3.73/4.0 – 2nd GPA 4.0/4.0 – 3rd GPA 4.0/4.0 – Cumulative GPA 3.89/4.0

- Development of three-tier Web applications that utilize the MVC architecture using Java and C#.
- Implementation of web apps that incorporate MERN, AWS and Flutter.
- Design, modeling, implementation, and maintenance of databases using Oracle, MySQL, MariaDB, Postgres DB, Mongo DB and DB2.
- Structured approach to program design, development testing and debugging of computer programs.
- Utilization of project management tools and software to manage a project (Agile Development/Waterfall Development).
- Analysis of problem situations to define the specifications of a system based on requirements.
- Development of mobile applications for Android and iOS.

2002-2003

Microsoft Partner Academy Courses (Training), Microsoft

- Vision 2 Results (Soft Skills Training).
- Project Management.
- Security.
- Systems Architecture and Networking.
- MSF (Microsoft Solutions Framework).
- Technical Writing.

1996-1999

Bachelor of Business Administration, 6th of October University in 6th of October City, Egypt

- Specialization: Finance - Minor: Economics - Degree: Very Good - Graduation Project: Distinct

SKILLS

- Languages/Frameworks/Databases /Cloud- Java SE/EE, JSP, Spring Boot, Python, JavaScript, RxJs, GSAP, MeteorJS, BunJS, C#/ASP.Net, Go Lang, MERN (MongoDB, ExpressJs, ReactJS, NodeJS), Gatsby, Vue.JS, Svelte, SolidJS, jQuery, Typescript, ES6, PHP, Laravel, React-native, , Flutter, Dart, AWS, Android Dev, IOS Dev, HTML, CSS, Oracle DB, DB2, MariaDB, PostgreSQL and MSSQL.
- Familiar with many libraries and plugins associated with the forementioned languages and frameworks.
- Familiar with many IDEs such as Visual Studio .Net, VSCode, Atom, Sublime, Eclipse, IntelliJ, Idle, PyCharm, DBeaver, Android Studio and many more
- Have a strong understanding of three tier architecture under many platforms and environments
- Strong grasp over Object-oriented design and functional programming
- Cloud computing, Rest API, Websockets
- Git and bash
- Team leader and a team player
- Problem solver and strong pattern recognizer
- I enjoy learning new things and can assimilate new ideas very quickly
- High attention to detail
- Worked in Agile/Scrum environments
- Fluent in English written and spoken holding an IELTS score of 8/9
- Firm grasp of software design before implementation
- Excellent Multitasker
- Minimal supervision